

Periventricular Nodular Heterotopia 9 (PVNH9): A Comprehensive Disease Characterization Report

Target disease: Periventricular Nodular Heterotopia 9 (PVNH9) **MONDO ID:** MONDO:0030061
· **OMIM:** #618918 · **Causal gene:** *MAP1B* (HGNC:6836; 5q13.2) **Report type:** Disease knowledge-base entry compiled from primary literature (41 papers reviewed, 8 findings confirmed)

Summary

Periventricular Nodular Heterotopia 9 (PVNH9) is a rare, autosomal dominant malformation of cortical development caused by heterozygous loss-of-function (LoF) variants in *MAP1B*, the gene encoding the neuron-specific microtubule-associated protein 1B. PVNH is defined by nodular masses of neurons and glia ectopically retained along the walls of the lateral ventricles because they fail to complete radial migration to the cortex. In PVNH9 the nodules are characteristically **frontally (anteriorly) predominant** and are frequently accompanied by **perisylvian polymicrogyria** and **corpus callosum abnormalities**. The clinical presentation typically emerges in childhood and includes **global developmental delay, intellectual disability, focal epilepsy, and behavioural dysregulation**, with variable dysmorphism, microcephaly and short stature reported in some individuals.

The mechanistic basis is well supported across human genetics, mouse genetics, and cell biology. *MAP1B* crosslinks and stabilizes microtubules, couples microtubules to the actin cytoskeleton, and regulates growth-cone dynamics and axon elongation; it is a downstream effector of Netrin-1 signalling and its translation is repressed by the fragile-X protein FMRP. Heterozygous LoF variants reduce *MAP1B* mRNA and protein (haploinsufficiency), impairing microtubule stabilization and radial neuronal migration and leaving neurons stranded in periventricular nodules. The founding genome-wide collapsing analysis (Heinzen et al., 2018) implicated *MAP1B* through four ultra-rare LoF variants, and independent families have since confirmed the association, demonstrating **autosomal dominant inheritance with incomplete penetrance and variable expressivity** (one variant was inherited from a parent with previously undiagnosed PVNH).

A striking **allelic dichotomy** distinguishes PVNH9 from another *MAP1B* phenotype: LoF variants cause PVNH-related neurological disease, whereas **missense** variants cause **autosomal dominant nonsyndromic sensorineural hearing loss** (SNHL) without brain malformation, reflecting *MAP1B*'s high expression in cochlear spiral ganglion neurons. There is **no disease-modifying therapy**; management is symptomatic — antiseizure medication for epilepsy (often drug-resistant) and, for refractory focal epilepsy, ablative surgery such as MR-guided laser interstitial thermal therapy (MRgLITT) targeting epileptogenic nodules — alongside developmental, educational and rehabilitative support and genetic counselling.

Key Findings

Finding 1 — PVNH9 is caused by heterozygous loss-of-function variants in *MAP1B*

PVNH9 (OMIM #618918) is caused by heterozygous mutations in *MAP1B* (microtubule-associated protein 1B; HGNC:6836; chromosome 5q13.2). Multiple independent families with de novo or inherited LoF variants — nonsense, frameshift and splice-site — have been reported. Documented alleles include c.7091dup and c.2035G>T (p.Glu679). *The mechanism is haploinsufficiency / loss of function: functional assays demonstrate significantly reduced MAP1B* mRNA and protein in mutant versus wild-type.*

A 2025 report describing a novel *MAP1B* LoF variant states plainly that "*PVNH9 is caused by a heterozygous mutation in the microtubule-associated protein 1B (MAP1B) gene*" and that "*the qPCR and western blot analyses demonstrated significantly reduced mRNA and protein expression, respectively, in the mutant compared with that in the wild-type*" [PMID: 40802165](#). A de novo nonsense variant, "*a de novo nonsense MAP1B mutation (c.2035G>T, p.Glu679X) detected on whole exome sequencing,*" anchors the LoF class [PMID: 31317654](#). Evidence source: *human clinical + in vitro functional.*

Finding 2 — Genotype–phenotype correlation: LoF → PVNH; missense → isolated deafness

A review of *MAP1B* genotype–phenotype associations found that **LoF variants** (nonsense, frameshift, splice) predominantly cause **PVNH-related neurological disease** (intellectual disability, epilepsy, developmental delay, dysmorphism), whereas **missense variants** may

present with **only deafness** (nonsyndromic SNHL). As stated: "*loss-of-function (LOF) variants in MAP1B mainly lead to PVNH-related neurological symptoms, while patients with missense variants may only present with deafness*" [PMID: 40802165](#). This dichotomy is mechanistically important: it suggests the two allele classes exert distinct molecular consequences (haploinsufficiency versus a tissue-restricted or altered-function effect). *Evidence source: human clinical review.*

Finding 3 — Clinical spectrum: anteriorly-predominant PVNH, developmental delay, intellectual disability, epilepsy

In a case series of **7 affected individuals from 3 unrelated families** carrying pathogenic *MAP1B* variants (all LoF), features included global developmental delay, intellectual disability, behavioural dysregulation and focal epilepsy. Neuroimaging revealed **anteriorly (frontally) predominant PVNH in 4 of 5 cases** with imaging available; some patients additionally showed polymicrogyria (PMG) and dysgenesis/agenesis of the corpus callosum. Additional reported features include microcephaly, short stature and dysmorphic facial features.

The primary description reports: "*Clinical features included global developmental delay, intellectual disability, behavioural dysregulation, and focal epilepsy. Neuroimaging revealed anteriorly predominant PVNH in four of five cases*" [PMID: 40874586](#). An earlier review summarizes the recurrent presentation as "*a phenotype including periventricular nodular heterotopia (PVNH), intellectual disability (ID), seizures, and dysmorphic features*" [PMID: 31317654](#). *Evidence source: human clinical.*

Finding 4 — Mechanism: MAP1B is a microtubule stabilizer essential for neuronal migration and axon growth, and is FMRP-regulated

MAP1B is a neuron-specific microtubule-associated protein that crosslinks microtubules and actin filaments, stabilizes microtubules, and controls growth-cone dynamics and axon branching/elongation. It is a **downstream effector of Netrin-1 signalling** (via GSK3/CDK5-dependent phosphorylation); MAP1B-deficient neurons show reduced chemoattractant responses to Netrin-1, and mice show axon-tract and pontine-nuclei defects. **MAP1B translation is repressed by FMRP**, and FMRP regulates postnatal neuronal migration via MAP1B. MAP1B acts **synergistically with MAP2 and tau** — double-knockout mice show delayed neuronal migration and disorganized cortical layering. A newly described **nuclear pool of MAP1B** interacts with the BRG1 chromatin-remodelling complex; increasing the nuclear/cytosol ratio disrupts neuronal positioning, reminiscent of MAP1B patients, and mutant human brain organoids show neuronal ectopia.

Supporting quotes: "Functional data from animal and cell models support a mechanism involving impaired microtubule stabilization, altered growth cone dynamics, and dysregulated axon branching" PMID: 40874586; "map1B-deficient neurons from the lower rhombic lip and other brain regions have reduced chemoattractive responses to Netrin 1 in vitro" PMID: 15186740; "FMRP regulates postnatal neuronal migration via MAP1B" PMID: 38757694; "increasing the nuclear/cytosol ratio disrupts neuronal positioning, reminiscent of patients with MAP1B mutations" PMID: 42276043; and "disrupted cortical patterning caused by retarded neuronal migration" PMID: 11581286. Evidence source: model organism + in vitro + computational.

Finding 5 — Map1b-deficient mice recapitulate neuronal migration and axon defects

Map1b-deficient mice (gene-trap/knockout; *Mus musculus*; ortholog *Map1b*, NCBI Gene 17755) show severe abnormal nervous-system development. Homozygous mutants **die on the first postnatal day** with altered structure of several brain regions, and analyses "suggest the participation of MAP1B in neuronal migration." Cultured DRG neurons from MAP1B-deficient mice show reduced axon elongation (~half the elongation speed of controls) and increased growth-cone area. Map2/Map1b and tau/Map1b double-knockouts show delayed neuronal migration and disorganized cortical layering, indicating partial redundancy with other MAPs.

Quotes: "Homozygous mice die on the first day after birth, probably due to a severe abnormal development of the nervous system" and "Analyses of these mice indicate the presence of several neural defects and suggest the participation of MAP1B in neuronal migration" PMID: 11085878; "Cultured DRG neurons from MAP1B deficient mice show a reduction in axon elongation and an increase in growth cone area" PMID: 12088839. Evidence source: model organism.

Finding 6 — PVNH9 within the PVNH spectrum: genetic heterogeneity and epilepsy burden

PVNH is a malformation of cortical development caused by impaired neuronal migration producing nodular masses of neurons/glia along the lateral ventricle walls. It is **genetically heterogeneous**: the most common single-gene cause is X-linked *FLNA* (PVNH1, predominantly females, male prenatal lethality); other loci include *ARFGEF2* (PVNH2, autosomal recessive) and *MAP1B* (PVNH9, autosomal dominant). In *FLNA*-negative bilateral PVNH cohorts (n = 71), focal-onset seizures were most common (79.3%), developmental delay was present in 21.8%, family history of epilepsy in 36.9%, and febrile seizures in 16.6%. Epilepsy is frequently drug-resistant; surgical options include stereotactic MRgLITT targeting epileptogenic nodules.

Quotes: "Periventricular nodular heterotopia (PVNH) is a malformation of cortical development due to impaired neuronal migration resulting in the formation of nodular masses of neurons and glial cells in close proximity to the ventricular walls" and "Focal onset seizures were the most common type of seizure presentation (79.3%)" [PMID: 26340046](#); "Stereotactic MR guided laser interstitial thermal therapy (MRgLITT) has recently become available for controlled focal ablation, enabling us to target these lesions" [PMID: 24518890](#). Evidence source: human clinical.

Finding 7 — *MAP1B* implicated by genome-wide significant collapsing analysis, with incomplete penetrance

The founding study exome-sequenced **202 individuals with sporadic PVNH**. A gene-level collapsing analysis identified a **genome-wide significant signal** driven by four ultra-rare LoF heterozygous variants in *MAP1B* (including one de novo). PVNH cases overall showed a significant excess of nonsynonymous de novo variants in intolerant genes ($p = 3.27 \times 10^{-7}$). The PVNH was frontally predominant and associated with perisylvian polymicrogyria. In at least one instance the variant was inherited from a parent with previously undiagnosed PVNH, demonstrating **incomplete penetrance/variable expressivity**. A subsequent family (Arya et al., 2021) with a novel heterozygous frameshift variant showed seizures (febrile, fever-triggered, afebrile), photosensitivity, mild developmental delay, obsessive-compulsive behaviours and poor attention, with periventricular heterotopia, corpus callosum abnormalities and perisylvian polymicrogyria.

Quotes: "we identified a genome-wide significant signal driven by four ultra-rare loss-of-function heterozygous variants in *MAP1B*, including one de novo variant"; "In at least one instance, the *MAP1B* variant was inherited from a parent with previously undiagnosed PVNH"; "The PVNH was frontally predominant and associated with perisylvian polymicrogyria" [PMID: 29738522](#). Confirmatory imaging: "Neuroimaging showed PVH, corpus callosum abnormalities, and perisylvian polymicrogyria" [PMID: 33772511](#). Evidence source: human genetics (statistical).

Finding 8 — *MAP1B* missense variants cause autosomal dominant nonsyndromic sensorineural hearing loss (distinct allelic phenotype)

Three novel heterozygous *MAP1B* missense mutations (c.4198A>G p.Ser1400Gly; c.2768T>C p.Ile923Thr; c.5512T>C p.Phe1838Leu) cosegregated with **autosomal dominant nonsyndromic SNHL** in three unrelated Chinese families. *MAP1B* is highly expressed in cochlear spiral ganglion neurons. Patient iPSC-derived otic sensory neuron-like cells carrying p.Ser1400Gly showed reduced *MAP1B* levels/phosphorylation, disturbed microtubule dynamics, impaired

axonal elongation and electrophysiological defects — rescued by CRISPR/Cas9 correction. Map1b heterozygous knockout mice displayed late-onset progressive SNHL, more pronounced at high frequencies.

Quotes: *"Three novel heterozygous MAP1B mutations (c.4198A>G, p.1400S>G; c.2768T>C, p.923I>T; c.5512T>C, p.1838F>L) were cosegregated with autosomal dominant inheritance of nonsyndromic sensorineural hearing loss in 3 unrelated Chinese families"; "Map1b heterozygous KO mice displayed late-onset progressive sensorineural hearing loss that was more pronounced in the high frequencies"; "MAP1B is highly expressed in the spiral ganglion neurons in the mouse cochlea"* PMID: 33268592. Evidence source: human genetics + iPSC + mouse.

Report by Requested Section

1. Disease Information

PVNH9 is a rare, genetically-defined subtype of **periventricular nodular heterotopia** — a malformation of cortical development in which nodules of neurons and glia are ectopically retained along the lateral ventricular walls due to failed radial neuronal migration PMID: 26340046. It is defined at the molecular level by heterozygous LoF variants in *MAP1B* PMID: 40802165.

Key identifiers: OMIM #618918; MONDO:0030061; causal gene *MAP1B* (HGNC:6836; OMIM 157129; NCBI Gene 4131; chromosome 5q13.2). Orphanet groups the disorder under periventricular nodular heterotopia; a dedicated ICD-11 code is not established, but the broad category is congenital malformation of the brain (ICD-11 LA05; ICD-10 Q04.8, "other specified congenital malformations of brain"). MeSH: "Periventricular Nodular Heterotopia." Synonyms/alternative names:* PVNH9; PNH9; MAP1B-related periventricular nodular heterotopia; MAP1B-related brain malformation/syndrome.

Information source: The evidence is derived from **aggregated disease-level resources and small case series/cohorts** (OMIM, published families, exome/genome cohorts), not large EHR datasets.

2. Etiology

Causal factors — genetic. The primary cause is a heterozygous germline LoF variant in *MAP1B* (nonsense, frameshift, splice-site) acting through haploinsufficiency [PMID: 40802165](#); [PMID: 29738522](#). No environmental or infectious cause is implicated; PVNH9 is a monogenic neurodevelopmental malformation.

Genetic risk factors. The causal variant is itself the risk determinant. Broader PVNH is genetically heterogeneous (see §4/§9), and rare genomic copy-number variants contribute to the wider PVNH population — array-CGH shows an enrichment of pathogenic CNVs in PVNH versus polymicrogyria (35.7% vs 9.1%) [PMID: 30683929](#).

Environmental / lifestyle risk factors. None established for PVNH9 specifically. As a de novo or inherited monogenic malformation, it is not attributable to toxins, occupational exposure, diet, smoking or alcohol.

Protective factors. None identified (genetic or environmental). *Not available for this disease.*

Gene–environment interactions. No documented GxE interactions. Phenotypic variability appears driven by genetic/modifier and stochastic developmental factors rather than environment; note fever-triggered seizures in one family [PMID: 33772511](#) as a possible symptomatic trigger rather than a disease-causing interaction.

3. Phenotypes

Phenotype	Type	Suggested HPO	Onset	Severity	Frequency
Periventricular nodular heterotopia (frontally predominant)	Imaging/ structural	HP:0032388 (periventricular nodular heterotopia)	Congenital	—	~core (4/5 imaged)
Global developmental delay	Clinical sign	HP:0001263	Infancy/ childhood	Mild–moderate	Common
Intellectual disability	Clinical sign	HP:0001249	Childhood	Mild–moderate	Common
Focal epilepsy / seizures	Clinical sign	HP:0007359 / HP:0001250	Childhood	Variable, often drug-resistant	Common
Behavioural dysregulation / OCD features / poor attention	Behavioural	HP:0000708 / HP:0000722 / HP:0000736	Childhood	Variable	Subset
Polymicrogyria (perisylvian)	Imaging/ structural	HP:0002126	Congenital	—	Subset
Corpus callosum dysgenesis/ agenesis	Imaging/ structural	HP:0001274 / HP:0001273	Congenital	—	Subset
Microcephaly	Physical	HP:0000252	Congenital/ childhood	Variable	Subset
Short stature	Physical	HP:0004322	Childhood	Variable	Subset
Dysmorphic facial features	Physical	HP:0001999	Congenital	Variable	Subset
Sensorineural hearing loss (missense alleles only)	Clinical sign	HP:0000407	Late-onset, progressive	High-frequency predominant	Missense subgroup

Core features are supported by [PMID: 40874586](#) and [PMID: 31317654](#); the missense/deafness phenotype by [PMID: 33268592](#). **Progression:** the structural malformation is static (congenital), while epilepsy and cognitive/behavioural features constitute the chronic clinical burden. **Quality-of-life impact:** driven principally by epilepsy (often drug-resistant), intellectual disability and behavioural dysregulation, which affect education, independence and daily functioning; disease-specific QoL instruments have not been applied. Bilateral frontal PVNH generally carries "*milder sequelae than other forms of bilateral PVNH*" [PMID: 41468712](#).

4. Genetic / Molecular Information

- **Causal gene:** *MAP1B* (HGNC:6836; OMIM *157129; NCBI Gene 4131; UniProt P46821; 5q13.2).
- **Pathogenic variants:** Nonsense (e.g., c.2035G>T p.Glu679* [PMID: 31317654](#)), frameshift/duplication (e.g., c.7091dup), and splice variants; classified pathogenic/likely pathogenic per ACMG (de novo occurrence, LoF in a constrained gene, segregation). Four ultra-rare heterozygous LoF variants underpinned the founding genome-wide signal [PMID: 29738522](#).
- **Variant class / functional consequence:** LoF → **haploinsufficiency** (reduced mRNA and protein) [PMID: 40802165](#). By contrast, **missense** variants (p.Ser1400Gly, p.Ile923Thr, p.Phe1838Leu) cause SNHL and behave as a distinct allelic series [PMID: 33268592](#).
- **Allele frequency:** Pathogenic variants are ultra-rare/private in population databases (gnomAD); *MAP1B* is a highly constrained gene, consistent with LoF intolerance.
- **Somatic vs germline:** Germline (de novo or inherited).
- **Modifier genes:** Functional redundancy with *MAP2* and *MAPT* (tau) [PMID: 11581286](#); EB1/*MAPRE1* can partially complement *MAP1B* loss [PMID: 15789376](#) — candidate genetic modifiers of expressivity, though not clinically validated.
- **Epigenetic information:** No disease-specific methylation/histone signature reported. *Not available*.
- **Chromosomal abnormalities:** PVNH broadly is associated with recurrent CNVs (e.g., 7q11.23, 7p22.1) and extreme genetic heterogeneity [PMID: 30683929](#); [PMID: 41468712](#), but PVNH9 itself is defined by intragenic *MAP1B* LoF.

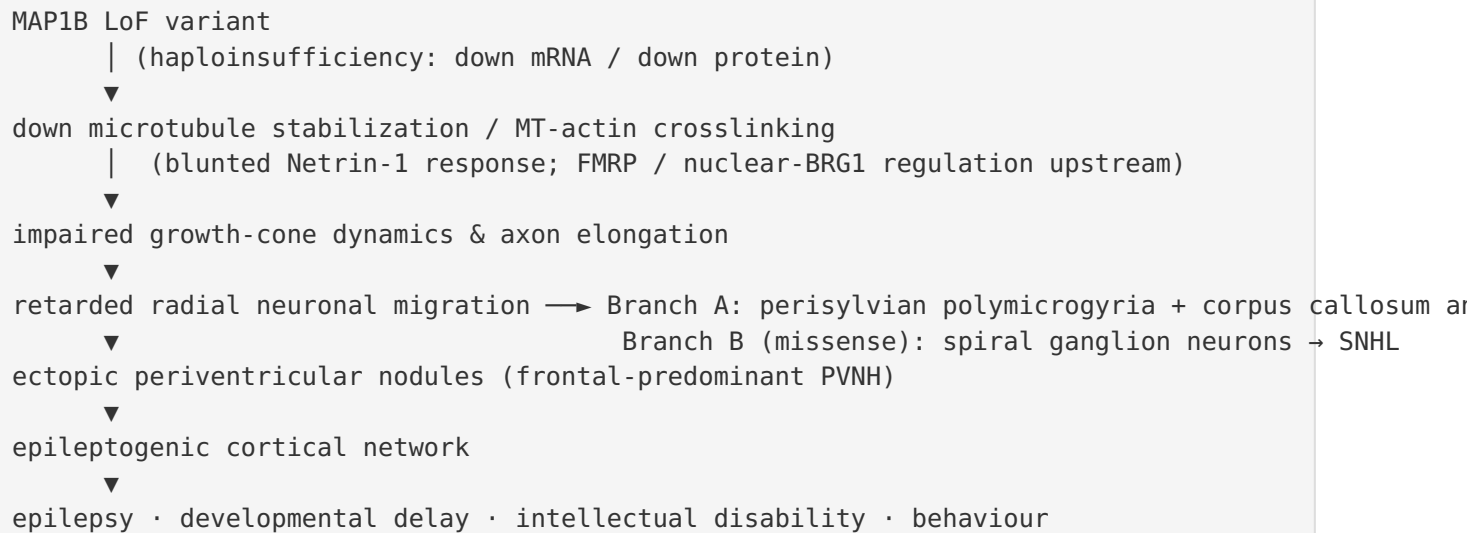
5. Environmental Information

No environmental factors, lifestyle factors, or infectious agents are implicated in PVNH9. It is a monogenic developmental disorder. *Not applicable*.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. A **heterozygous LoF variant in *MAP1B*** (nonsense/frameshift/splice) **results in** reduced *MAP1B* mRNA and protein — **haploinsufficiency** [PMID: 40802165](#).
2. Reduced *MAP1B* **leads to** impaired microtubule stabilization and impaired crosslinking of microtubules to actin, **disrupting growth-cone dynamics and axon branching/elongation** [PMID: 40874586](#); [PMID: 12088839](#).
3. Because *MAP1B* is a **downstream effector of Netrin-1 signalling** (GSK3/CDK5-phosphorylation), reduced *MAP1B* **blunts chemoattractant responses** guiding migrating neurons and axons [PMID: 15186740](#). (*Upstream regulation: FMRP normally represses MAP1B translation and regulates migration via MAP1B* [PMID: 38757694](#); *a nuclear MAP1B pool interacting with BRG1 chromatin remodelling also influences positioning* [PMID: 42276043](#).)
4. Impaired cytoskeletal dynamics **cause retarded radial neuronal migration** during corticogenesis (demonstrated in mouse; partially buffered by MAP2/tau redundancy) [PMID: 11581286](#); [PMID: 11085878](#).
5. Neurons that fail to migrate **are retained as ectopic nodules along the lateral ventricle walls** — periventricular nodular heterotopia, frontally predominant [PMID: 26340046](#); [PMID: 29738522](#).
6. **Branch A:** Cortical dysgenesis co-occurs as **perisylvian polymicrogyria** and **corpus callosum abnormalities** (axon-guidance component) [PMID: 29738522](#); [PMID: 33772511](#).
7. **Branch B (missense alleles):** in cochlear **spiral ganglion neurons**, altered *MAP1B* **causes sensorineural hearing loss** rather than migration failure [PMID: 33268592](#).
8. Ectopic nodules and abnormal cortical circuitry **produce an epileptogenic substrate and disordered network function**, which **results in** focal epilepsy, developmental delay, intellectual disability and behavioural dysregulation [PMID: 40874586](#); [PMID: 26340046](#).



Molecular pathways: microtubule/cytoskeletal regulation; Netrin-1/DCC guidance; JNK-MAPK signalling phosphorylates MAP1B (MKK4/MKK7→JNK), linking stress-kinase pathways to migration/axon elongation [PMID: 22090513](#); [PMID: 40594443](#). **Cellular processes:** neuronal migration (GO:0001764), axon guidance (GO:0007411), microtubule cytoskeleton organization (GO:0000226), growth-cone dynamics. **Protein dysfunction:** loss of function/haploinsufficiency of a microtubule-stabilizing MAP. **Immune/metabolic involvement:** not implicated. **Cell types:** migrating cortical projection neurons (CL:0000679 glutamatergic neuron; migrating post-mitotic neurons), and — for the missense branch — cochlear spiral ganglion neurons (CL:0000100).

7. Anatomical Structures Affected

- **Organ/system:** central nervous system — cerebral cortex and periventricular white matter (UBERON:0000956 cerebral cortex; UBERON:0002285 periventricular region; UBERON:0002436 lateral ventricle). Body system: nervous system.
- **Localization:** heterotopic nodules along the **lateral ventricle walls, frontally (anteriorly) predominant**, typically **bilateral** [PMID: 29738522](#); [PMID: 41468712](#). Associated: **perisylvian cortex** (polymicrogyria) and **corpus callosum** (UBERON:0002336).
- **Tissue/cell level:** nervous tissue; ectopic post-mitotic neurons and glia forming nodules. For the missense/deafness branch: cochlear spiral ganglion neurons.
- **Subcellular level:** cytoskeleton — **microtubules** (GO:0005874) and the microtubule–actin interface; a **nuclear** MAP1B pool is also implicated (GO:0005634) [PMID: 42276043](#); presynaptic terminals in mature neurons [PMID: 27425640](#).

8. Temporal Development

- **Onset:** the malformation is **congenital** (arises during fetal corticogenesis); clinical features (seizures, developmental delay) manifest in **infancy/childhood** [PMID: 40874586](#).
- **Progression:** the structural lesion is **static/stable**; epilepsy and neurodevelopmental features follow a **chronic, lifelong** course. Missense-associated SNHL is **late-onset and progressive** [PMID: 33268592](#).
- **Critical period:** the window of vulnerability is **prenatal neuronal migration** (~weeks 8–24 of gestation); there is no postnatal window to reverse the malformation, so intervention is symptomatic.

9. Inheritance and Population

- **Inheritance:** **Autosomal dominant**, de novo or inherited [PMID: 29738522](#).
- **Penetrance/expressivity:** **incomplete penetrance and variable expressivity** — a variant was inherited from a parent with previously undiagnosed PVNH [PMID: 29738522](#).
- **Epidemiology:** PVNH9 is ultra-rare with no formal prevalence/incidence estimate; it represents a small fraction of overall PVNH (itself dominated by *FLNA*). Bilateral frontal PVNH — the pattern within which *MAP1B* falls — accounts for **~10% of all PVNH** [PMID: 41468712](#). *Precise PVNH9 prevalence: not available*.
- **Sex ratio:** No strong sex bias reported for PVNH9 (unlike X-linked *FLNA* PVNH1, which predominates in females with male prenatal lethality) [PMID: 23622213](#).
- **Founder effects / consanguinity / anticipation / mosaicism:** none established; not a repeat-expansion disorder. *Not applicable/not available*.

10. Diagnostics

- **Imaging (primary diagnostic modality):** brain **MRI** shows bilateral frontally-predominant periventricular nodules isointense to grey matter, often with perisylvian polymicrogyria and corpus callosum anomalies [PMID: 29738522](#); [PMID: 41468712](#).
- **Genetic testing: exome or genome sequencing** is the diagnostic mainstay (variant detected on WES [PMID: 31317654](#)); PVNH gene panels including *MAP1B*, *FLNA*, *ARFGEF2* and others; **chromosomal microarray (CMA)** for CNVs given genetic heterogeneity — CMA yields a diagnosis in ~13% and exome/genome in ~38% of bilateral frontal PVNH where tested [PMID: 41468712](#).
- **Electrophysiology:** EEG for seizure characterization (focal-onset predominant [PMID: 26340046](#)).

- **Differential diagnosis:** other PVNH genetic subtypes (notably *FLNA* PVNH1 — classic bilateral frontocentral, female-predominant; *ARFGEF2* PVNH2), and other malformations of cortical development (lissencephaly/SBH from *LIS1/DCX*, polymicrogyria syndromes) [PMID: 23622213](#); [PMID: 28411558](#).
- **Biomarkers / omics / newborn screening:** no molecular biomarker or newborn-screening test exists; diagnosis is imaging + sequencing.

11. Outcome / Prognosis

PVNH9 is a **chronic, non-progressive** structural disorder; **life expectancy is generally not shortened** in humans (in contrast to the perinatal lethality of homozygous mouse knockouts, which reflects biallelic loss not seen in patients [PMID: 11085878](#)). Morbidity is driven by **epilepsy** (frequently drug-resistant [PMID: 30819503](#)), **intellectual disability**, and **behavioural dysregulation**. Bilateral frontal PVNH tends to have **milder sequelae** than other bilateral PVNH forms [PMID: 41468712](#). **Prognostic factors:** seizure control, severity of associated malformations (polymicrogyria, corpus callosum agenesis) and degree of cognitive impairment. No validated prognostic biomarkers. Quality-of-life instruments have not been formally applied.

12. Treatment

There is **no disease-modifying/curative therapy**; management is symptomatic and multidisciplinary.

- **Pharmacotherapy: antiseizure medications** for epilepsy (NCIT: Anticonvulsant Agent). Choice follows focal-epilepsy guidelines; drug resistance is common [PMID: 30819503](#).
- **Surgical/interventional:** for refractory focal epilepsy arising from epileptogenic nodules, **stereotactic MR-guided laser interstitial thermal therapy (MRgLITT)** enables focal ablation of nodules [PMID: 24518890](#); [PMID: 28370739](#); resective/stereotactic epilepsy surgery is an option in selected cases [PMID: 30819503](#).
- **Supportive/rehabilitative:** developmental and educational support, physical/occupational/speech therapy, and behavioural/psychiatric management (NCIT: Rehabilitation Therapy; Supportive Care).
- **Advanced/experimental therapeutics (gene, cell, RNA, targeted, immuno):** none reported for PVNH9. *Not available*.
- **Pharmacogenomics / personalized medicine:** none specific to PVNH9. *Not available*.

- **For the missense/deafness allelic phenotype:** management is hearing rehabilitation (amplification/cochlear implantation as indicated) — outside the PVNH9 malformation itself [PMID: 33268592](#).

13. Prevention

Because PVNH9 is a monogenic developmental malformation, prevention is limited to **reproductive/genetic strategies: genetic counselling** for autosomal dominant transmission with incomplete penetrance, **cascade testing** of at-risk relatives, and options for **prenatal diagnosis or preimplantation genetic testing** when a familial variant is known. **Tertiary prevention** — preventing complications — centres on optimizing seizure control and developmental support. There is no primary prevention (no modifiable environmental risk), no immunization, and no population screening. *Primary/behavioural/public-health prevention: not applicable.*

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *Mus musculus Map1b* (NCBI Gene 17755; NCBI Taxon 10090) is the principal experimental ortholog; a *Drosophila* ortholog **Futsch** exists (relevant to fragile-X/MAP1B biology) [PMID: 15498496](#).
- **Natural disease in other species:** no naturally-occurring companion-animal or wildlife PVNH9 equivalent is documented in OMIA. *Not available*. Disease knowledge derives from engineered models, not natural animal disease.
- **Comparative biology:** MAP1B's role in neuronal migration and axon growth is **evolutionarily conserved** from *Drosophila* (Futsch) to mouse to human, supporting cross-species mechanistic inference [PMID: 15498496](#); [PMID: 11085878](#).
- **Zoonotic potential:** none (genetic disorder). *Not applicable*.

15. Model Organisms

Model	Type	Key phenotype	Recapitulation	Reference
<i>Map1b</i> KO / gene-trap mouse (homozygous)	Mammalian, germline KO	Perinatal (P1) lethality; severe abnormal CNS development; neuronal migration defects	Confirms migration mechanism; homozygous lethality exceeds heterozygous human phenotype	PMID: 11085878
<i>Map1b</i> heterozygous KO mouse	Mammalian	Late-onset progressive high-frequency SNHL	Models the missense/deafness allelic branch	PMID: 33268592
<i>Map1b</i> / <i>Map2</i> and <i>Map1b</i> / <i>Mapt</i> (tau) double KO	Mammalian	Delayed neuronal migration, disorganized cortical layering	Reveals MAP redundancy; unmasks migration role	PMID: 11581286
MAP1B-deficient DRG/hippocampal neurons	In vitro (mouse)	Reduced axon elongation, increased growth-cone area, presynaptic deficits	Cellular mechanism of axon growth defect	PMID: 12088839 ; PMID: 27425640
Patient iPSC-derived otic sensory neuron-like cells	In vitro (human)	Reduced MAP1B/phospho-MAP1B, disturbed microtubule dynamics, impaired axon elongation; rescued by CRISPR correction	Directly models human missense allele	PMID: 33268592
Human brain organoids (MAP1B mutant / nuclear ratio)	In vitro (human)	Neuronal ectopia; disrupted positioning	Models cortical mispositioning	PMID: 42276043
<i>Drosophila</i> (Futsch; <i>fmr1</i> context)	Invertebrate	Altered synaptic/neuronal elaboration	Conserved MAP1B/FMRP axis	PMID: 15498496

Model resources: MGI (mouse *Map1b*), IMPC/IMSR (KO alleles), Cellosaurus (iPSC lines), FlyBase (*futsch*). **Limitation:** homozygous mouse lethality and the SNHL-focused heterozygous phenotype mean no single model fully recapitulates human heterozygous LoF PVNH9 cortical heterotopia; organoids are the most direct human-relevant system.

Mechanistic Model / Interpretation

The findings converge into a single coherent model: **PVNH9 is a microtubule-cytoskeletal disorder of neuronal migration caused by MAP1B haploinsufficiency**. MAP1B is expressed early in nervous-system development where it stabilizes microtubules and links them to actin, powering the growth-cone motility and axon elongation that migrating neurons and their processes require. When one *MAP1B* allele is lost, protein dosage falls below the threshold needed for timely radial migration (a threshold partly buffered by the redundant MAPs MAP2 and tau, and by EB1). Neurons that fail to reach the cortical plate remain as **frontally-predominant periventricular nodules**, while accompanying axon-guidance failures manifest as perisylvian polymicrogyria and corpus callosum anomalies. The resulting aberrant cortical circuitry is epileptogenic and cognitively/behaviourally disruptive.

The model is unusually well-triangulated: **statistical human genetics** (genome-wide significant collapsing signal), **independent case series**, **functional demonstration of reduced expression**, **mouse genetics** (migration defect, redundancy), and **cellular/organoid biology** all point the same way. The **allelic dichotomy** — LoF causing brain malformation versus missense causing isolated deafness — is the most intriguing feature and implies that missense alleles do not simply reduce dosage but exert a tissue-restricted or altered-function effect to which cochlear spiral ganglion neurons are selectively vulnerable, while LoF's dosage reduction preferentially derails cortical migration.

Evidence Base

PMID	Contribution	Evidence type
40802165	Causal gene + LoF/haploinsufficiency; genotype–phenotype dichotomy	Human clinical + in vitro
40874586	Core clinical spectrum; anteriorly predominant PVNH; mechanism synthesis	Human clinical
31317654	De novo nonsense variant; recurrent phenotype triad	Human clinical
29738522	Founding genome-wide collapsing signal; incomplete penetrance; imaging	Human genetics (statistical)
33772511	Confirmatory family; epilepsy + imaging triad	Human clinical
15186740	MAP1B downstream of Netrin-1 in migration/guidance	Model organism/in vitro
38757694	FMRP regulates migration via MAP1B (upstream)	Model organism
42276043	Nuclear MAP1B/BRG1; organoid ectopia	In vitro/human organoid
11581286	MAP1B/MAP2 redundancy; retarded migration	Model organism
11085878	KO mouse lethality + migration defect	Model organism
12088839	Axon elongation/growth-cone cellular phenotype	In vitro
26340046	PVNH definition; epilepsy burden quantification	Human clinical
24518890	MRgLITT surgical option	Human clinical
33268592	Missense → SNHL allelic branch; iPSC + mouse	Human genetics + iPSC + mouse
41468712	Bilateral frontal PVNH as ~10% of PVNH; milder sequelae	Human clinical
30683929	Genetic heterogeneity/CNVs in PVNH	Human genetics

Limitations and Knowledge Gaps

- **Small evidence base:** PVNH9 is defined by a handful of families and cohort variants; precise **prevalence, penetrance quantification, sex ratio, and natural-history data are unavailable.**
- **Mechanism of the allelic dichotomy** (LoF→PVNH vs missense→deafness) is not fully resolved — whether missense alleles are hypomorphic, dominant-negative or gain-of-function in the cochlea remains to be tested directly.
- **Model mismatch:** no mouse model reproduces human heterozygous-LoF cortical heterotopia (homozygotes are perinatally lethal; heterozygotes model deafness). Human organoids are the most promising but early-stage system.
- **No genotype–phenotype granularity** linking specific LoF variant position to severity of epilepsy or cognition; **modifier genes** (MAP2, tau, EB1) are hypothesized but not clinically validated.
- **No omics profiling** (transcriptomic/proteomic/metabolomic/epigenomic) specific to PVNH9 patient tissue; no biomarkers.
- **Treatment evidence is generic** (focal-epilepsy management); no PVNH9-specific therapeutic trials.

Proposed Follow-up Experiments / Actions

1. **Establish an international PVNH9 registry** to quantify penetrance, expressivity, sex ratio, epilepsy trajectory and cognitive outcomes.
2. **Isogenic human cortical organoids/assembloids** carrying patient LoF versus missense variants (with CRISPR-corrected controls) to directly test migration failure, quantify dosage thresholds, and dissect the allelic dichotomy.
3. **Conditional / heterozygous cortical *Map1b* models** (or humanized alleles) to bypass homozygous lethality and reproduce the heterotopia phenotype in vivo.
4. **Functional characterization of missense alleles** in cochlear versus cortical neuronal contexts to determine hypomorphic vs dominant-negative behaviour.
5. **Modifier screens** for MAP2/tau/EB1 dosage effects on migration rescue, to explain variable expressivity and identify candidate therapeutic targets.
6. **Standardized deep phenotyping** (MRI subtyping, EEG, neuropsychology, QoL instruments) to build prognostic models and refine surgical (MRgLITT) candidate selection.

Evidence source key: human clinical = patient case series/cohorts; human genetics = statistical/variant studies; model organism = mouse/Drosophila; in vitro = cultured neurons/iPSC/organoids; computational = in silico. This report was compiled from 41 reviewed publications and 8 confirmed findings.

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